ASSIGNMENT-2

Foundations of Cryptography

Course: BCA | Subject: Cryptography

Q1. List the four fundamental cryptographic goals/objectives.

The four fundamental objectives are confidentiality, integrity, authentication, and non-repudiation.

Q2. Define confidentiality as a cryptographic goal with one example.

Confidentiality means that information is accessible only to authorized people. For example, encryption protects online banking details so that someone intercepting the network traffic cannot read them.

Q3. Explain confidentiality, integrity, authentication and non-repudiation with real-world scenarios. Why are all four important together?

Confidentiality protects information from being read by unauthorized people; for example, encryption protects a private message. Integrity ensures that data has not been changed without permission; for example, a message authentication code can detect an altered network message. Authentication verifies the identity of a user or the source of a message; for example, a server certificate helps a browser verify a website. Non-repudiation provides evidence that a particular party performed an action, often through a digital signature. All four are important because a message may be private but altered, unchanged but sent by an impostor, or properly authenticated but later disputed. Together, suitable mechanisms help provide secure and trustworthy communication.

Q4. What is the difference between a substitution cipher and a transposition cipher? Give one example of each.

A substitution cipher replaces each letter or group of letters with other symbols, such as the Caesar cipher. A transposition cipher keeps the same letters but changes their positions, such as the Rail Fence cipher.

Q5. Discuss the evolution from classical ciphers to modern standards (DES → AES, RSA, SHA), including milestones and motivations for change.

Classical ciphers, such as Caesar and Vigenère, used manual substitution and transposition and could be attacked using language patterns and statistical analysis. Mechanical systems such as Enigma increased complexity, but wartime cryptanalysis showed the need for stronger mathematical methods. DES was adopted in 1977 as a computer-era symmetric block cipher, but its effective 56-bit key became too short against modern brute-force attacks. AES was selected as its successor and standardized in 2001, offering a 128-bit block and 128-, 192- or 256-bit keys. RSA, a public-key algorithm introduced in the 1970s, supports operations such as key establishment and digital signatures and helps address key-distribution challenges. SHA is a family of cryptographic hash functions used for integrity checks and as part of digital-signature systems; it is not an encryption algorithm. These developments were motivated by growing computing power, networked communication, the need for scalable key distribution and stronger security.

Q6. What is a key in cryptography, and why is key management important?

A key is a value used by a cryptographic algorithm to encrypt, decrypt, sign or verify data. Key management is important because keys must be generated, shared, stored, rotated and revoked securely; a compromised key can undermine the protection provided by a strong algorithm.

Q7. Explain how cryptographic techniques protect data during transmission, prevent unauthorized access and detect tampering in networks.

During transmission, protocols such as TLS encrypt data to protect confidentiality and use authenticated cryptographic mechanisms to detect changes. Certificates and authentication methods help verify the identity of servers and, where configured, users. VPN technologies such as IPsec can protect traffic between networks or devices. Passwords should be stored using a suitable salted password-hashing scheme rather than reversible encryption. Digital signatures can verify that a message or software update was signed by the holder of a private key. Together, encryption, authentication, message authentication codes, secure hashes and digital signatures help protect network communication against eavesdropping, impersonation and tampering.

Q8. Differentiate between a passive attack and an active attack, with one example of each.

A passive attack observes communication without changing it, such as eavesdropping on unencrypted network traffic. An active attack changes, injects or disrupts communication, such as modifying a message in transit or replaying a captured authentication message.

Q9. Describe how cryptography evolved to counter new threats, including the move from symmetric-only systems to public-key cryptography and its solution to key distribution.

Early systems mainly used symmetric techniques, in which both parties needed the same secret key. This could be efficient, but the sender and receiver had to share the key securely beforehand; distributing separate keys for many users became difficult at scale. Advances in mathematics and computing led to public-key cryptography, publicly introduced by Diffie and Hellman in 1976, followed by RSA in 1977. Public-key systems use a public key and a related private key. This allows users to establish shared secrets or verify digital signatures without sending the private key over the network. In modern protocols, asymmetric methods are often combined with fast symmetric encryption: the parties authenticate and establish session keys, then use symmetric algorithms to protect large amounts of data. New threats and increased computing power continue to drive stronger key sizes, better protocols and regular algorithm updates.

Q10. Give one example each of a modern symmetric and asymmetric algorithm.

A modern symmetric algorithm is AES, which uses a shared secret key. An asymmetric algorithm is RSA, which uses a public/private key pair. Elliptic-curve cryptography is another widely used asymmetric approach.

Q11. Classify cryptography into classical and modern cryptography and explain the transition to computer-based algorithms.

Classical cryptography includes manual substitution and transposition methods, such as Caesar, Vigenère and Rail Fence ciphers. These methods often relied on letter patterns and were vulnerable to frequency analysis. Mechanical devices such as Enigma automated more complex transformations, especially during World War II. Modern cryptography uses mathematical algorithms designed for computers, including symmetric block ciphers such as DES and AES, public-key systems such as RSA and elliptic-curve cryptography, and hash functions such as SHA-256. Important milestones include Shannon's theoretical work, the public-key ideas published in 1976, the adoption of DES in 1977 and the standardization of AES in 2001. Modern systems also use protocols such as TLS and IPsec to apply cryptography in real networks. The transition reflects the need to protect large volumes of digital data, manage keys across many users and withstand powerful computational attacks.

References

1. William Stallings, Cryptography and Network Security: Principles and Practice.

2. NIST publications on DES, AES and Secure Hash Standards.

3. Whitfield Diffie and Martin E. Hellman, New Directions in Cryptography, 1976.